Department of Computer Science and Engineering
Fall 2025
Course Code: IMG 401
Course Title: Industry and Operational Management
Assignment on Quality Management
Due Date: 2nd June 2026

Case Study: Boeing 737 MAX MCAS Failure

The Boeing 737 MAX was developed to compete with Airbus's fuel-efficient A320neo aircraft. Instead of designing a completely new airplane, Boeing modified its existing 737 platform and introduced larger engines. Due to changes in engine placement, the aircraft had a tendency to pitch upward during certain flight conditions.

To address this issue, Boeing introduced a software system called **MCAS (Maneuvering Characteristics Augmentation System)**. MCAS automatically pushed the aircraft nose downward when it detected a high angle of attack.

However, the system relied heavily on data from a single sensor. Faulty sensor readings caused MCAS to activate incorrectly during flights. Two major crashes occurred:

- Lion Air Flight 610 (2018)
- Ethiopian Airlines Flight 302 (2019)

A total of 346 people died, and the Boeing 737 MAX fleet was grounded worldwide.

Investigations later revealed concerns regarding:

- Inadequate software testing
- Weak risk assessment
- Poor communication with pilots
- Schedule pressure to compete with Airbus
- Insufficient quality assurance and certification processes

The case is now widely studied as a major failure in quality management, operational decision-making, and safety-critical software engineering.

Assignment Task

Write a report answering the following questions based on the Boeing 737 MAX MCAS failure case study.

(a) In your opinion, what were the major causes behind the Boeing 737 MAX crisis?

(b) Do you think this was mainly:
  - a technical failure,
  - a management failure,
  - or both?

  Explain your reasoning.

(c) Which quality management principles were neglected by Boeing?

(d) Which quality management tools can be applied to analyze this case? Explain briefly how they could help.

Examples include:

- Fishbone (Ishikawa) Diagram
- 5 Whys Analysis
- Failure Mode and Effects Analysis (FMEA)
- Pareto Analysis
- PDCA Cycle
- Risk Matrix

(e) Why are testing, redundancy, and communication especially important in safety-critical systems?

(f) What lessons can software and hardware companies learn from this case regarding quality management?

Rules and Guidelines

- The report must be written in your own words in a report format not like question and answer.
- The submission should contain at least 350 words.
- Relevant quality management concepts and tools should be clearly discussed.
- The assignment must be handwritten and submitted online as scanned copy.
- Copying from other students is strictly prohibited.
- Plagiarism or use of AI-generated content without proper understanding may result in marks deduction.
- Mention all references or sources used at the end of the report.
- Late submissions may not be accepted without valid justification.